

An effective-ODE (averaging-theory) drift bound for one round of FederatedAveraging

Mini-study extending McMahan et al. (2017), arXiv:1602.05629

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Abstract

For one round of FederatedAveraging (FedAvg) we idealize each client's u_k local full-batch gradient steps as the gradient flow of its local objective run for continuous time $T_k = \eta u_k$. We prove that the server map—the weight-averaged composition of these per-client flows—agrees with the gradient flow of the *global* objective f to second order in the time horizon, with the leading discrepancy equal to a curvature–gradient covariance across clients:

$$\left\| \sum_k \beta_k \Phi_k^T(w) - \Phi_f^T(w) \right\| = \frac{T^2}{2} \sigma_{HG} + O(T^3), \quad \sigma_{HG} = \|\text{Cov}_k(H_k, g_k)\|,$$

with an explicit cubic remainder under a bounded-third-derivative assumption. This is exact for all local-step counts and shows the per-round drift depends on (E, B, η) only through the single invariant $T = \eta E n_k / B$. The heuristic $\Delta = \eta^2 \text{Cov}_k(H_k, g_k)$ of the parent study's deep dive is recovered as the $E = 2$, $B = \infty$ special case (where $T = 2\eta$), with the correct prefactor $T^2/2$ replacing the frozen η^2 .

1 Setup

There are K clients with local objectives $F_k : \mathbb{R}^d \rightarrow \mathbb{R}$, weights $\beta_k \geq 0$, $\sum_k \beta_k = 1$, and global objective

$$f = \sum_{k=1}^K \beta_k F_k.$$

We write the gradient and Hessian at the broadcast point w as $g_k = \nabla F_k(w)$, $H_k = \nabla^2 F_k(w)$, $g = \nabla f(w) = \sum_k \beta_k g_k$, $H = \nabla^2 f(w) = \sum_k \beta_k H_k$. (We treat $w \in \mathbb{R}^d$; the matrix-parameter softmax model of the prototype is the case $d = \text{dim}$ with the Euclidean/Frobenius inner product, to which everything applies verbatim.)

Assumption 1 (Smoothness). There is a closed ball $\mathcal{B} \subset \mathbb{R}^d$ containing w and a constant $\rho < \infty$ such that each $F_k \in C^3(\mathcal{B})$ with $\sup_{v \in \mathcal{B}} \|\nabla^3 F_k(v)\| \leq \rho$ and $\sup_{v \in \mathcal{B}} \|\nabla F_k(v)\| \leq G$, $\sup_{v \in \mathcal{B}} \|\nabla^2 F_k(v)\| \leq L$.

For $T \geq 0$ small enough that all flow trajectories below remain in $\mathcal{B}$, define the gradient-flow maps

$$\Phi_k^T(w) = \phi_k^T, \quad \dot{\phi}_k^s = -\nabla F_k(\phi_k^s), \quad \phi_k^0 = w; \quad \Phi_f^T(w) = \psi^T, \quad \dot{\psi}^s = -\nabla f(\psi^s), \quad \psi^0 = w. \quad (1)$$

The FedAvg *server map* is the weighted average of per-client flow maps,

$$\mathcal{S}^T(w) := \sum_{k=1}^K \beta_k \Phi_k^T(w). \quad (2)$$

The horizon $T = \eta u_k$ is the continuous-time image of $u_k = En_k/B$ discrete local steps of size η ; see Lemma 2 for the discrete link.

2 Lie–Taylor expansion of a gradient flow

Lemma 1 (Third-order flow expansion). *Under Assumption 1, for $0 \leq T \leq T_0$ (with T_0 chosen so the trajectory stays in $\mathcal{B}$),*

$$\Phi_k^T(w) = w - Tg_k + \frac{T^2}{2} H_k g_k + R_k(T), \quad \|R_k(T)\| \leq \frac{T^3}{6} C_3, \quad (3)$$

where $C_3 := \rho G^2 + L^2 G$ bounds the third s -derivative of ϕ_k^s on $\mathcal{B}$. The same holds for Φ_f^T with (g_k, H_k) replaced by (g, H) and the same constant (since f is a convex combination of the F_k).

Proof. Let $V = -\nabla F_k$, so $\dot{\phi}^s = V(\phi^s)$. Then $\frac{d}{ds}\phi^s = V(\phi^s)$ and, differentiating again, $\frac{d^2}{ds^2}\phi^s = DV(\phi^s)V(\phi^s)$, where $DV = -\nabla^2 F_k$. At $s = 0$: $\phi^0 = w$, $\dot{\phi}^0 = -g_k$, $\ddot{\phi}^0 = (-H_k)(-g_k) = H_k g_k$. Taylor's theorem with the Lagrange/integral remainder gives (3) with $R_k(T) = \frac{1}{6} \int_0^T (T-s)^2 \ddot{\phi}^s ds$ in the integral form, so $\|R_k(T)\| \leq \frac{T^3}{6} \sup_{s \leq T} \|\ddot{\phi}^s\|$. Differentiating a third time,

$$\ddot{\phi}^s = \frac{d}{ds} [DV(\phi^s)V(\phi^s)] = D^2V(\phi^s)(V(\phi^s), V(\phi^s)) + DV(\phi^s)DV(\phi^s)V(\phi^s),$$

and $\|D^2V\| = \|\nabla^3 F_k\| \leq \rho$, $\|DV\| = \|\nabla^2 F_k\| \leq L$, $\|V\| = \|\nabla F_k\| \leq G$ on $\mathcal{B}$, so $\|\ddot{\phi}^s\| \leq \rho G^2 + L^2 G = C_3$. The statement for Φ_f^T is identical because $\nabla^3 f = \sum_k \beta_k \nabla^3 F_k$ obeys the same bound by convexity of the norm and $\sum_k \beta_k = 1$. $\square$

3 Main result

Theorem 1 (Effective-ODE drift bound). *Let Assumption 1 hold and suppose all clients share the same horizon $T_k \equiv T$ (equal effective time; e.g. equal client sizes). Then for $0 \leq T \leq T_0$,*

$$\mathcal{S}^T(w) - \Phi_f^T(w) = \frac{T^2}{2} \Gamma + R(T), \quad \Gamma := \sum_k \beta_k H_k g_k - Hg = \text{Cov}_k(H_k, g_k), \quad (4)$$

with $\|R(T)\| \leq \frac{T^3}{3} C_3$. Consequently, writing $\sigma_{HG} := \|\Gamma\|$,

$$\left| \|\mathcal{S}^T(w) - \Phi_f^T(w)\| - \frac{T^2}{2} \sigma_{HG} \right| \leq \frac{T^3}{3} C_3, \quad \text{i.e.} \quad \|\mathcal{S}^T(w) - \Phi_f^T(w)\| = \frac{T^2}{2} \sigma_{HG} + O(T^3), \quad (5)$$

and the leading-order drift direction is exactly $\Gamma / \|\Gamma\|$.

Proof. Average (3) over k with weights β_k :

$$\mathcal{S}^T(w) = \sum_k \beta_k \Phi_k^T(w) = w - T \sum_k \beta_k g_k + \frac{T^2}{2} \sum_k \beta_k H_k g_k + \sum_k \beta_k R_k(T) = w - Tg + \frac{T^2}{2} \sum_k \beta_k H_k g_k + R_1(T),$$

with $\|R_1(T)\| = \|\sum_k \beta_k R_k(T)\| \leq \sum_k \beta_k \frac{T^3}{6} C_3 = \frac{T^3}{6} C_3$ (Jensen / triangle inequality, $\sum_k \beta_k = 1$). By Lemma 1 applied to f , $\Phi_f^T(w) = w - Tg + \frac{T^2}{2} Hg + R_2(T)$ with $\|R_2(T)\| \leq \frac{T^3}{6} C_3$. Subtracting, the $O(1)$ term w and the $O(T)$ term $-Tg$ cancel exactly, leaving

$$\mathcal{S}^T(w) - \Phi_f^T(w) = \frac{T^2}{2} \left(\sum_k \beta_k H_k g_k - Hg \right) + (R_1(T) - R_2(T)) = \frac{T^2}{2} \Gamma + R(T),$$

$\|R(T)\| \leq \|R_1\| + \|R_2\| \leq \frac{T^3}{3} C_3$. The covariance identity is the algebra

$$\Gamma = \sum_k \beta_k H_k g_k - \left(\sum_k \beta_k H_k \right) \left(\sum_k \beta_k g_k \right) = \mathbb{E}_k[H_k g_k] - \mathbb{E}_k[H_k] \mathbb{E}_k[g_k] = \text{Cov}_k(H_k, g_k),$$

where $\mathbb{E}_k$ is expectation under the probability weights β_k . Finally (5) is the reverse triangle inequality applied to (4). $\square$

Remark 1 (Unequal horizons). If client horizons differ, $T_k = \eta u_k$ with $u_k = En_k/B$, define the weighted mean horizon $\bar{T} = \sum_k \beta_k T_k$. Then $\mathcal{S}(w) - \Phi_f^{\bar{T}}(w) = -\sum_k \beta_k (T_k - \bar{T}) g_k + \frac{1}{2} \left(\sum_k \beta_k T_k^2 H_k g_k - \bar{T}^2 H g \right) + O(T^3)$. The new first-order term $\sum_k \beta_k (T_k - \bar{T}) g_k = \text{Cov}_k(T_k, g_k)$ is the *size-heterogeneity* drift; it vanishes when client sizes are equal ($T_k \equiv T$, recovering Theorem 1) or when one compares against the size-weighted reference $\sum_k \beta_k \Phi_f^{T_k}$. Either way the result is governed entirely by the horizons $\{T_k = \eta En_k/B\}$, never by (E, B, η) separately.

Lemma 2 (Discrete-to-flow consistency). *Let $\hat{\Phi}_k^T$ be the u_k -step explicit Euler discretization of (1) with step $\eta = T/u_k$ (i.e. literal full-batch local GD). Then $\|\hat{\Phi}_k^T(w) - \Phi_k^T(w)\| \leq \frac{T}{2} LG \eta = \frac{LGT^2}{2u_k} \rightarrow 0$ as $u_k \rightarrow \infty$ at fixed T . Hence the discrete FedAvg server map satisfies the bound (5) up to an additional $O(\eta) = O(T/u_k)$ term, which the prototype confirms vanishes as E grows at fixed T .*

Proof. Standard global-error bound for explicit Euler on a field with Jacobian bound L and second-derivative-of-solution bound $\|\ddot{\phi}\| \leq LG$: the global error is $\leq \frac{\eta}{2L}(e^{LT} - 1) \sup \|\ddot{\phi}\|$; for LT bounded this is $O(\eta)$ with constant $\leq \frac{1}{2} LG e^{LT} T$. Averaging over k preserves the bound. $\square$

4 Consequences

Effective outer ODE. Iterating Theorem 1, the server sequence $w_{t+1} = \Phi_f^T(w_t) - \frac{T^2}{2} \Gamma(w_t) + O(T^3)$ shadows, in the slow time $s = tT$ and to first order in T , the perturbed flow

$$\frac{dw}{ds} = -\nabla f(w) - \frac{T}{2} \text{Cov}_k(H_k, g_k) + O(T^2).$$

The heterogeneity field $\text{Cov}_k(H_k, g_k)$ is the continuous-time avatar of the deep-dive's Δ ; it vanishes when clients are statistically identical ($H_k \equiv H$ or $g_k \equiv g$), recovering exact centralized gradient flow.

Recovering and correcting the heuristic. For $E = 2$, $B = \infty$ we have $u_k = 2$, $T = 2\eta$, and Theorem 1 gives drift $\frac{(2\eta)^2}{2} \sigma_{HG} = 2\eta^2 \sigma_{HG}$ with the *same* object $\Gamma = \text{Cov}_k(H_k, g_k)$ as the two-step Taylor sketch. For general E the correct prefactor is $T^2/2 = (\eta u_k)^2/2$, not the frozen η^2 : the heuristic mispredicts the magnitude by a factor $u_k^2/4$ (e.g. $\approx 6\times$ at $E = 5$). The single invariant is $T = \eta En_k/B$.

A Softmax cross-entropy satisfies Assumption 1

For mean cross-entropy with softmax probabilities $p_i = \text{softmax}(x_i^\top W)$, the gradient is $\frac{1}{m} \sum_i x_i (p_i - y_i)^\top$ and the Hessian acts as $H[V] = \frac{1}{m} \sum_i x_i ((\text{diag}(p_i) - p_i p_i^\top)(x_i^\top V))^\top$ (used verbatim in the prototype's exact Hessian-vector product). All derivatives are bounded on any bounded W -set because $p \mapsto \text{softmax}$ is C^∞ with bounded derivatives ($\|p\| \leq 1$) and the data $\{x_i\}$ are fixed; hence Assumption 1 holds on any ball, with constants depending only on $\max_i \|x_i\|$.